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安徽大学纽约石溪学院数字逻辑实验报告

【实验名称】Application of Decoder

【实验目的】

1.Familiarize with the experiment box.

2.Master the logical functions and usage methods of medium-scale integrated circuit decoders.

3.Utilize decoders to design combinational logic circuits.

【实验原理】

1.The logical function of a variable decoder is to translate the input n-bit binary code into 2^n output variables. Each output variable corresponds to a unique set of input codes.

2.The 3-line to 8-line decoder 74LS138 is a typical variable decoder. When the input is a certain set of codes, only the corresponding output signal is at a valid level, while other outputs are invalid at a high level.

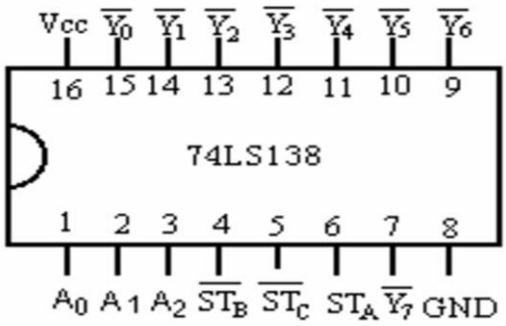

Fig 2- 1 Pinout Diagram of 3-to-8 Line Decoder 74LS138

ST_A	$\overline{ST_B}+\overline{ST_C}$	A_2	A_1	A_0	$\overline{Y_0}$	$\overline{Y_1}$	$\overline{Y_2}$	$\overline{Y_3}$	$\overline{Y_4}$	$\overline{Y_5}$	$\overline{Y_6}$	$\overline{Y_7}$
×	1	×	×	×	1	1	1	1	1	1	1	1
0	×	×	×	×	1	1	1	1	1	1	1	1
1	0	0	0	0	0	1	1	1	1	1	1	1
1	0	0	0	1	1	0	1	1	1	1	1	1
1	0	0	1	0	1	1	0	1	1	1	1	1
1	0	0	1	1	1	1	1	0	1	1	1	1
1	0	1	0	0	1	1	1	1	0	1	1	1
1	0	1	0	1	1	1	1	1	1	0	1	1
1	0	1	1	0	1	1	1	1	1	1	0	1
1	0	1	1	1	1	1	1	1	1	1	1	0

Table 2- 1 The truth table of 3-line to 8-line decoder

Apart from decoding, a decoder can also be used as a data distributor. If an input terminal of the enabling pin is used to input data information, the device becomes a data distributor.

With a decoder, it is also convenient to implement logical functions with multiple outputs.

【实验内容】

1. Test the logical function of the 3-line to 8-line decoder 74LS138 and record the experimental data.

2. Implement a pulse distributor using 74LS138. Connect a 1Hz continuous pulse signal to one of the enable control terminals STB/STC of 74LS138, and determine the selected channel based on the address input terminals. Change the logical states of the address input terminals sequentially (from 000 to 111), observe the changes at the output terminals, and conduct a detailed analysis.

3. Design and implement a 1-bit binary full subtractor circuit using 74LS138 and

gate circuits.

4. Design and implement a combinational logic circuit with a specific function using 74LS138 and gate circuits.

$$\begin{cases} Y_1 = \bar{B}\bar{C} + AB\bar{C} \\ Y_2 = AC \\ Y_3 = A\bar{B} + \bar{B}C \end{cases}$$

【实验分析及设计】

2. Implement a pulse distributor using 74LS138. (Combined with experiment I)

Input						***Output***							
ST_A	$\overline{ST_B}$	$\overline{ST_C}$	C	B	A	$\overline{Y_0}$	$\overline{Y_1}$	$\overline{Y_2}$	$\overline{Y_3}$	$\overline{Y_4}$	$\overline{Y_5}$	$\overline{Y_6}$	$\overline{Y_7}$
1	Impulse	0	0	0	0	Flashing	1	1	1	1	1	1	1
1	Impulse	0	0	0	1	1	Flashing	1	1	1	1	1	1
1	Impulse	0	0	1	0	1	1	Flashing	1	1	1	1	1
1	Impulse	0	0	1	1	1	1	1	Flashing	1	1	1	1
1	Impulse	0	1	0	0	1	1	1	1	Flashing	1	1	1
1	Impulse	0	1	0	1	1	1	1	1	1	Flashing	1	1
1	Impulse	0	1	1	0	1	1	1	1	1	1	Flashing	1
1	Impulse	0	1	1	1	1	1	1	1	1	1	1	Flashing

Table 2- 2 Data I

3. Design and implement a 1-bit binary full subtractor circuit using 74LS138 and gate circuits.

A_i	B_i	C_{i-1}	S_i	C_i
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

Table 2- 3 The truth table of 1-bit binary full subtractor

Thus, we have

$$S_i = \overline{A_i}\,\overline{B_i}C_{i-1} + \overline{A_i}B_i\overline{C_{i-1}} + A_i\overline{B_i}\,\overline{C_{i-1}} + A_iB_iC_{i-1} = A_i \oplus B_i \oplus C_{i-1}$$
$$C_i = \overline{A_i}\,\overline{B_i}C_{i-1} + \overline{A_i}B_i\overline{C_{i-1}} + \overline{A_i}B_iC_{i-1} + A_iB_iC_{i-1} = \overline{A_i}(B_i \oplus C_{i-1}) + B_iC_{i-1}$$
$$= \overline{\overline{\overline{A_i}(B_i \oplus C_{i-1})} \cdot \overline{B_iC_{i-1}}}$$

4. Design and implement a combinational logic circuit with a specific function using 74LS138 and gate circuits.

$$\begin{cases} Y_1 = \bar{B}\bar{C} + AB\bar{C} = A\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C} + AB\bar{C} = m_0 + m_4 + m_6 = \overline{\overline{Y_0} \cdot \overline{Y_4} \cdot \overline{Y_6}} \\ Y_2 = AC = ABC + A\bar{B}C = m_5 + m_7 = \overline{\overline{Y_5} \cdot \overline{Y_7}} \\ Y_3 = A\bar{B} + \bar{B}C = A\bar{B}C + \bar{A}\bar{B}C + A\bar{B}\bar{C} = m_1 + m_4 + m_5 = \overline{\overline{Y_1} \cdot \overline{Y_4} \cdot \overline{Y_5}} \end{cases}$$

【小结或讨论】